

Lecture - 3

Typical image processing pipeline



****** Transform: pixel values in some other, but equivalent form.

example: fouriers, wavelet, unity (no transform)

Special Processing

****** Direct manipulation of pixels (i.e. unity transform)

Types of Special Processing

- ① Point Processing
- ② Neighborhood Processing
- ③ Global Processing

Why Special Processing

- ① Enhancement $\rightarrow$ improved visual quality
- ② Improve recognition rate.

Mapping Function

****** $S = T(r)$

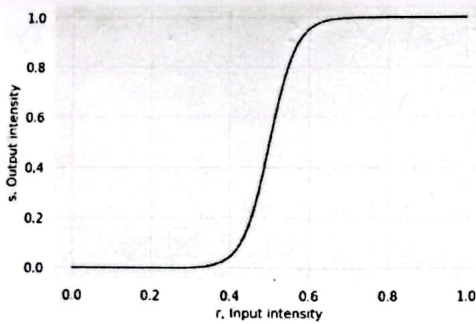
where,

S = output intensity

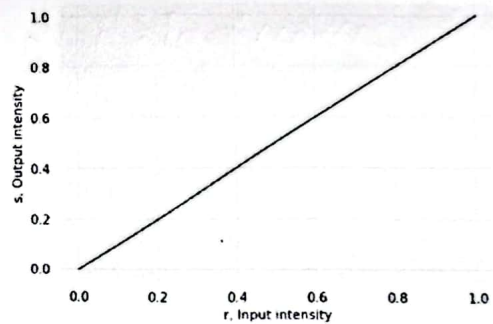
r = input intensity

$T[r]$ = mapping function

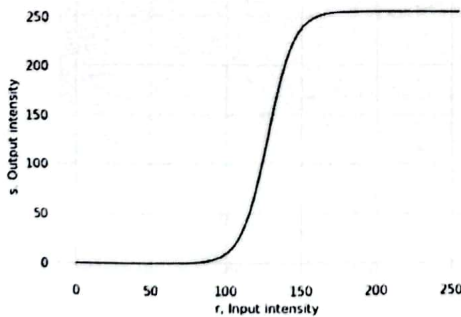
****** Example: Contrast stretching, Unity mapping.



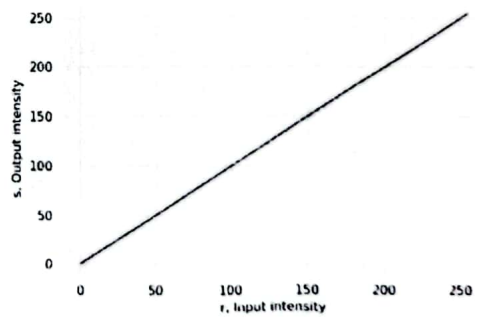
Contrast stretching



Unity mapping



Contrast stretching



Unity mapping

Arithmetic Operations

** Addition & Subtraction (Brightness Adjustment)

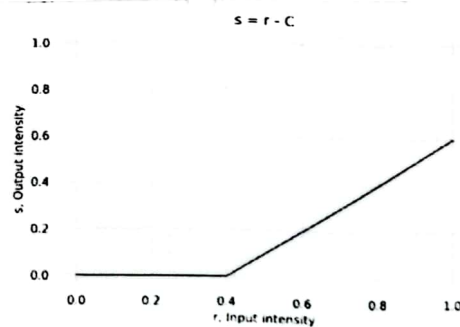
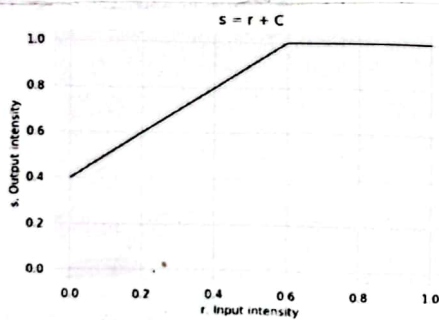
- $S = r \pm C$
- Have to be clipped between $[0, 1]$
- Increases (+) or decreases (-) the overall brightness.

↳ It does not change Contrast, when clipping not used.

contrast: Contrast = $I_b - I_d$

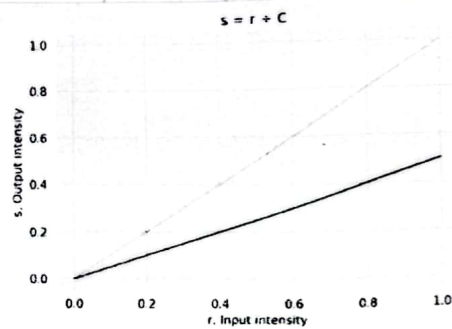
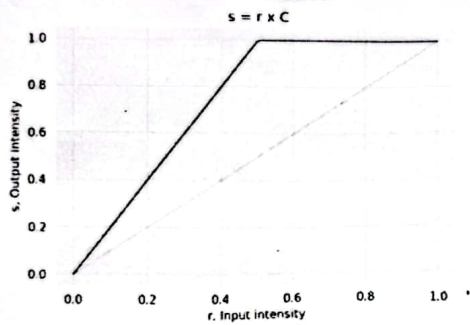
where, I_b = intensity of the brightest pixel

I_d = intensity of the darkest pixel



** Multiplication & Division (Brightness + Contrast) Adjustment

- $S = r \times C$ or $S = r \div C$
- $\text{np.clip}(r \times C, 0, 1)$ or $\text{np.clip}(r \div C, 0, 1)$
- It is combined with addition/subtraction for contrast adjustment.



A wide range near black/white

} input

↓
COMPRESSED
to
↓

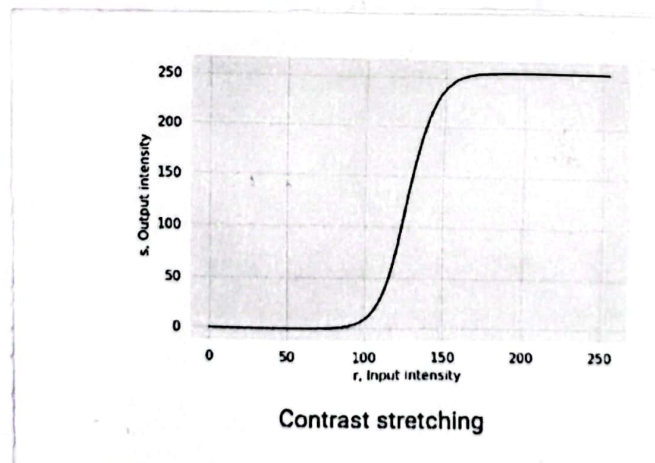
A corresponding narrow range

**

A small range near the middle

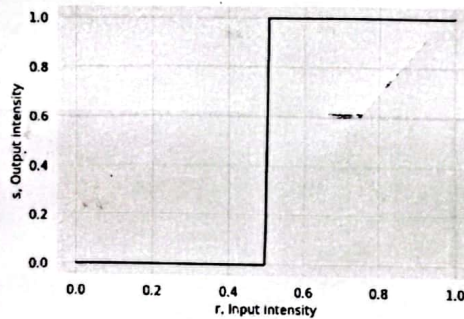
↓
Expanded
to
↓

A wide range



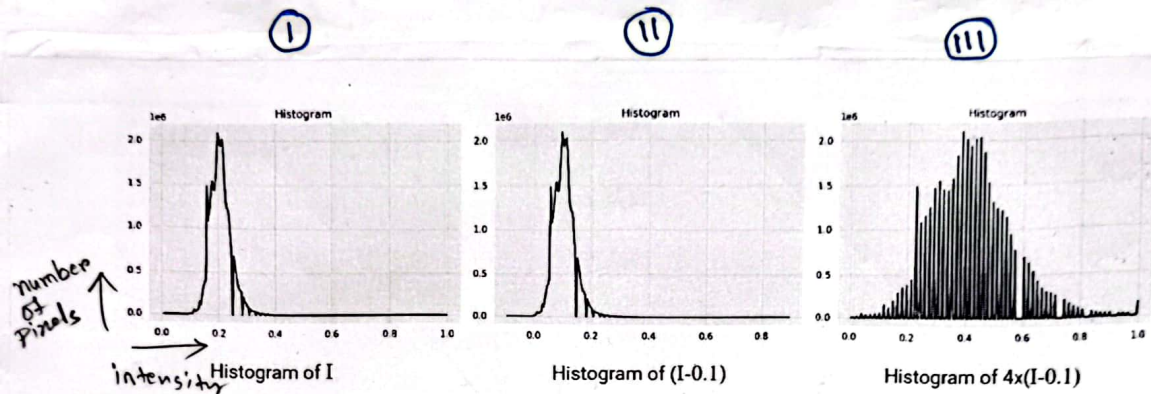
****** Limiting case of contrast stretching is thresholding.

→ Produces binary image



Thresholding

Contrast Stretching using Arithmetic Operations



⇒ Let's see the histogram of an image.
[Histogram ①]

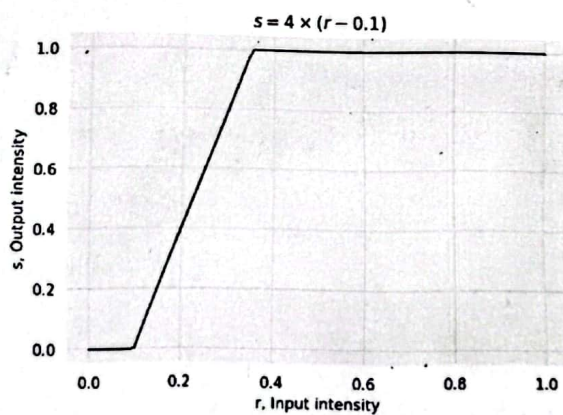
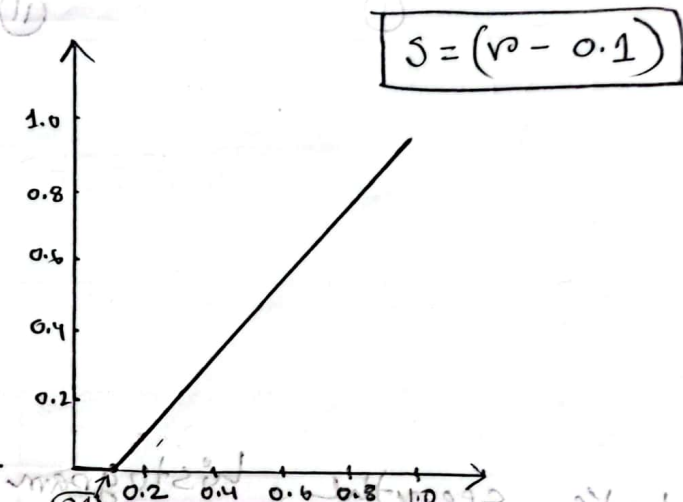
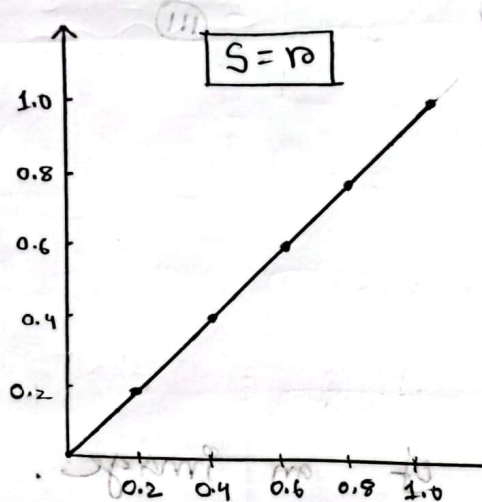
⇒ We can see that all the pixels are ranging between the intensity level of 0.1 to 0.35, which means this image has a very low contrast.

⇒ How can we improve?

If we can distribute the pixels that was between $[0.1, 0.35]$ to between $[0, 1]$, we can increase the contrast.

That is what has been done with some arithmetic operations (Histogram ② & ③)

Let's see the process in a mapping function.

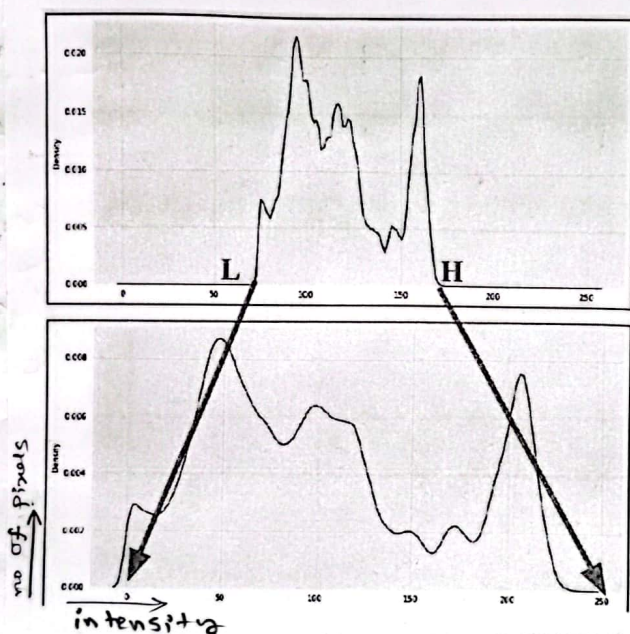


**

$$S = C_2 (r - C_1)$$

Here, $C_1 = L$

$$C_2 = \frac{1 - 0}{H - L} = \frac{255 - 0}{H - L}$$



L = lowest intensity of input image

H = highest intensity of input image

Contrast Stretching using Piecewise Linear Mapping

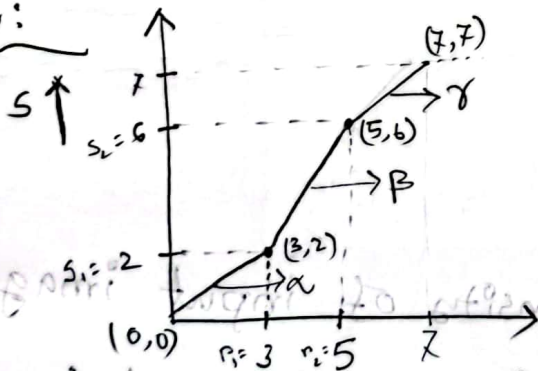
Question:

input image:

4	3	5	2
3	6	4	6
2	2	6	5
7	6	4	1

This is a 4x4 3-bit image. What is the output after you are done with contrast stretching. Given, $(r_1, s_1) = (3, 2)$
 $(r_2, s_2) = (5, 6)$

Solution:



$$\alpha = \frac{s_2 - s_1}{r_2 - r_1} = \frac{2 - 0}{3 - 0} = \frac{2}{3} = 0.67$$

$$\beta = \frac{s_2 - s_1}{r_2 - r_1} = \frac{6 - 2}{5 - 3} = \frac{4}{2} = 2$$

$$\gamma = \frac{s_2 - s_1}{r_2 - r_1} = \frac{7 - 6}{7 - 5} = \frac{1}{2} = 0.5$$

$$\text{Now, } S = \begin{cases} \alpha \cdot r & \text{if } 0 \leq r < 3 \\ \beta(r - r_1) + s_1 & \text{if } 3 \leq r < 5 \\ \gamma(r - r_2) + s_2 & \text{if } 5 \leq r \leq 7 \end{cases}$$

$$r_1 = 3, s_1 = 2$$

$$r_2 = 5, s_2 = 6$$

Now the output:

r	S
0	0
1	0
2	1
3	2
4	4
5	6
6	7
7	7

$$S = \alpha \cdot r = 0.67 \times 0 = 0 \approx 0$$

$$S = \alpha \cdot r = 0.67 \times 1 = 0.67 \approx 0$$

$$S = \alpha \cdot r = 0.67 \times 2 = 1.34 \approx 1$$

$$S = \beta(r - r_1) + s_1 = 2(3 - 3) + 2 = 2 \approx 2$$

$$S = \beta(r - r_1) + s_1 = 2(4 - 3) + 2 = 4 \approx 4$$

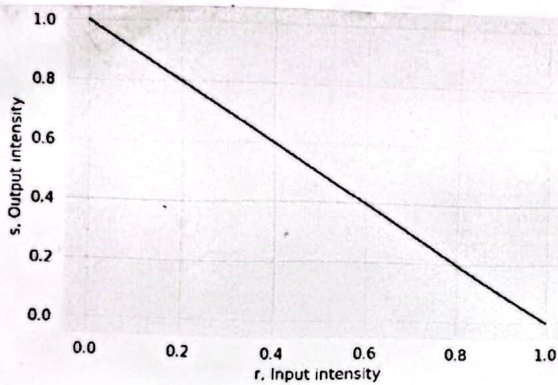
$$S = \gamma(r - r_2) + s_2 = 0.5(5 - 5) + 6 = 6 \approx 6$$

$$S = \gamma(r - r_2) + s_2 = 0.5(6 - 5) + 6 = 6.5 \approx 7$$

$$S = \gamma(r - r_2) + s_2 = 0.5(7 - 5) + 6 = 7 \approx 7$$

Some Intensity Transformation

Negative Image

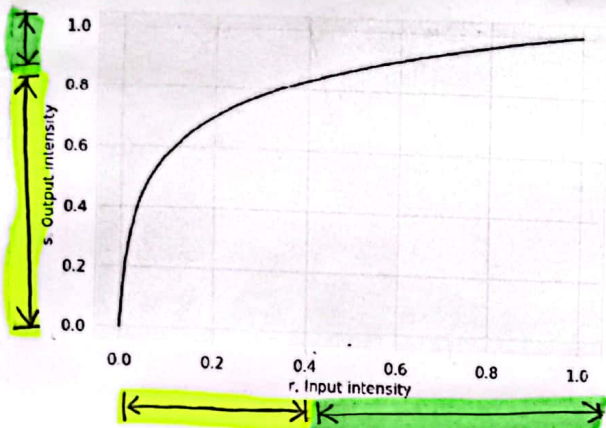


$$s = 1 - r$$

• Medical imaging

0	0
0	1
1	0
1	1
0	0
0	0
0	0
0	0

Log transform - Dynamic Range Compression

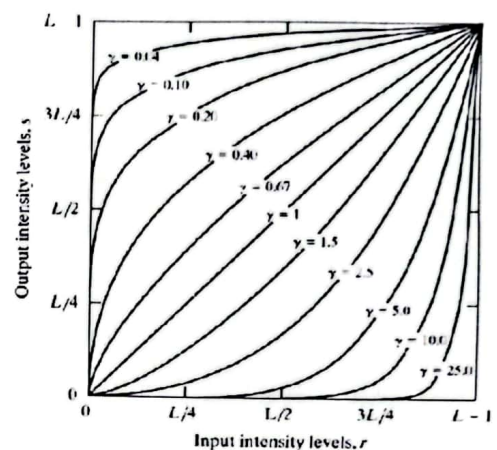
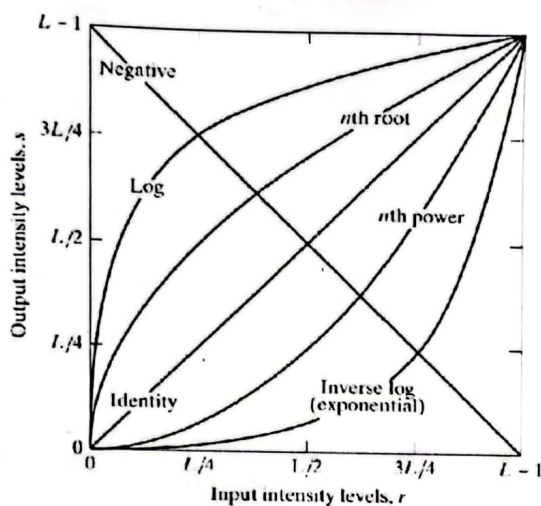


- $s = c \log(1+r)$; $r \geq 0$, $c \geq 0$

- Low range gray level expanded

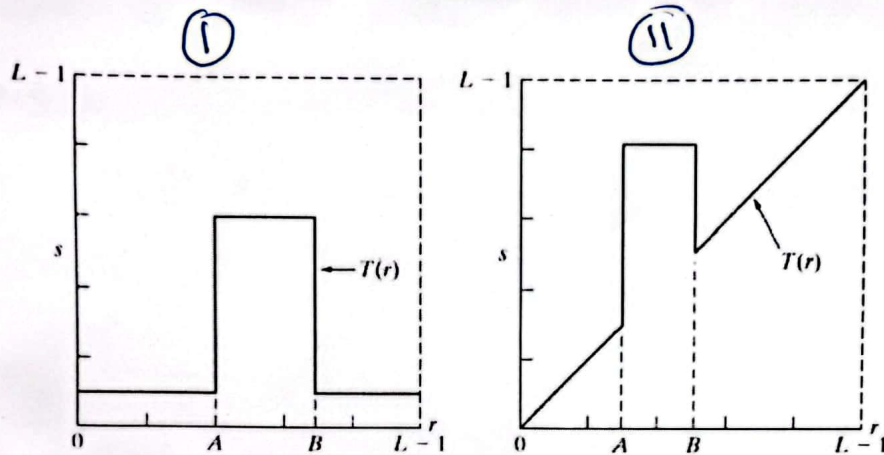
- High range levels compressed

Power Law Transformation



- $s = cr^\gamma$, $r \geq 0$, $c > 0$, $\gamma > 0$
- A generalized version of log transform.
- $\gamma > 1$ - Inverse log transformation, dynamic range expansion.
- $\gamma < 1$ - log transform, dynamic range compression.

Gray Level Slicing



- Increasing the intensity of a specific region.
- Rest of the regions either dark $\rightarrow$ (I) or unchanged $\rightarrow$ (II)

Lecture-4

Histogram

** for each gray level, how many pixels have the exact value.

$$\bullet h_r(r_k) = n_k \quad ; r = 0, 1, 2, 3, \dots, 255$$

Normalized Histogram

** for each gray level, what percentage of pixels have that exact value.

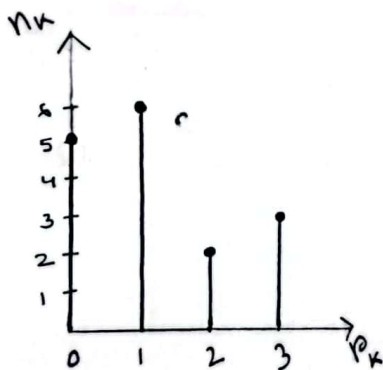
$$\bullet f_r(r_k) = \frac{n_k}{N_T} \quad ; N_T = H \times W$$

Example:

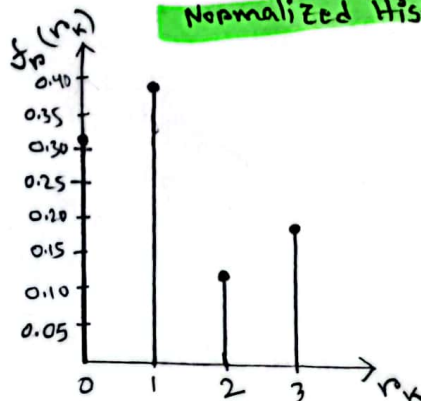
3	0	0	1
1	2	1	1
0	3	3	1
1	0	2	0

r_k	n_k	$f_r(r_k)$
0	5	$5/16 = 0.31$
1	6	$6/16 = 0.38$
2	2	$2/16 = 0.12$
3	3	$3/16 = 0.19$

Histogram



Normalized Histogram (PDF)



**** Normalized Histogram is also called**

Probability Density Function (PDF).

r_k	n_k	$f_n(r_k)$
0	5	0.31
1	6	0.38
2	2	0.12
3	3	0.19

31% pixels have intensity of level 0.

So, if we pick a random pixel there is a probability of 31% that the pixel will have the intensity level of 0.

**** Histograms are not unique.**

$$f_n(r_k) = \frac{n_k}{N}$$

Normalized Histogram

$$F_n(r_k) = \sum_{i=0}^k f_n(r_i)$$

Cumulative Histogram

for each gray level,
how many pixels have values $\leq$ that value

- $H_n(r_k) = n_k = \int_{-\infty}^{r_k} h_n(x) dx$; $r \in R$

- n_k is number of pixels with intensity $\leq r_k$

- $r \in R \rightarrow r$ can be any value, does not have to be discrete. e.g. 1.5, 2.3, ...

- $\int_{-\infty}^n h_n(x) dx \rightarrow$ integration of histogram = cumulative histogram

Normalized Cumulative Histogram

for each gray level,

what percentage of pixels have values $\leq$ that value.

- $F_n(r_k) = \frac{n_k}{N_T} = \int_{-\infty}^{r_k} f_n(x) dx$

Ex 1

Normalized Cumulative Histogram is also called **Cumulative Density Function (CDF)**

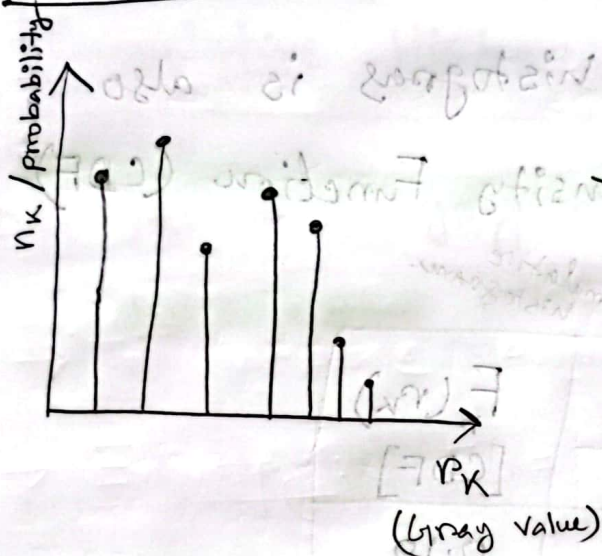
r_k	$h_n(r_k)$ → histogram	$f_n(r_k)$ [PDF]	$H_n(r_k)$ → Cumulative histogram.	$F_n(r_k)$ [CDF]
0	790	0.19 ($\frac{790}{4096}$)	790	0.19
1	1023	0.25	1813	
2	850	0.21	2663	
3	656	0.16	3319	0.81
4	329	0.08	3648	0.89
5	245	0.06	3893	0.95
6	122	0.03	4015	0.98
7	81	0.02	4096	1.0

$N_T = 64 \times 64 = 4096$

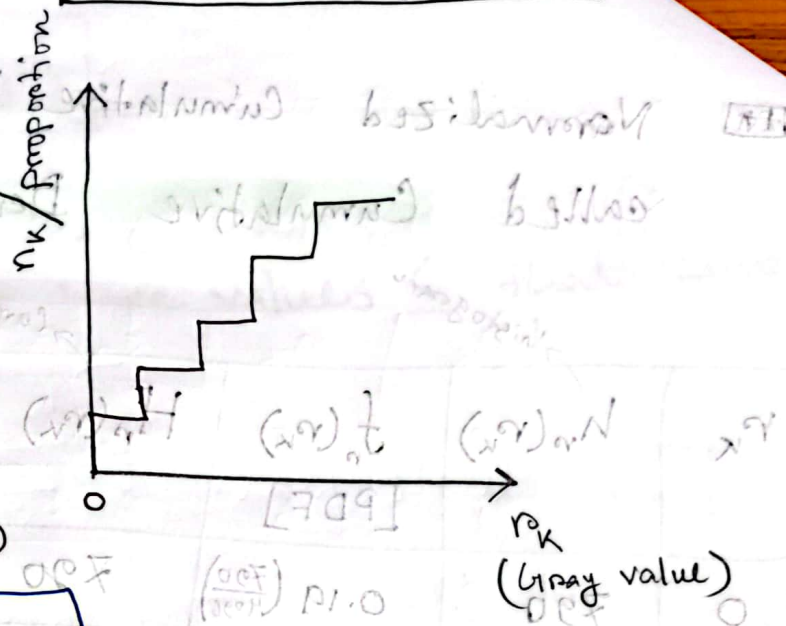
• Histogram is a statistical representation of data.

• The histogram is a statistical representation of data.

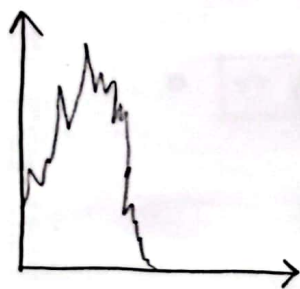
Histogram / PDF



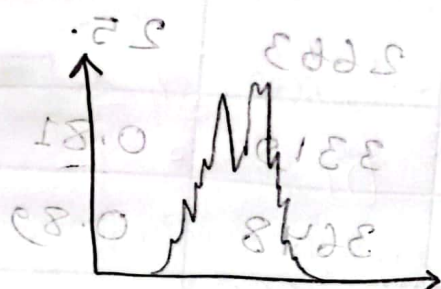
Cumulative Histogram / CDF



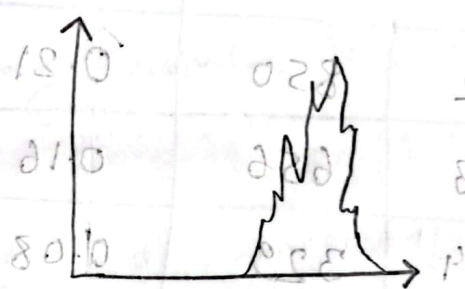
Examples of Histogram



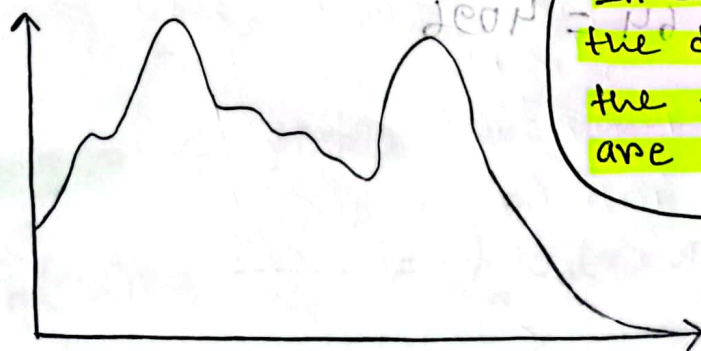
- Dark image
- Histogram concentrated in a low-range gray value
- Low contrast.



- Washed-out image
- Mid-range gray scale values
- Low Contrast.



- Bright image
- High-range gray scale values.
- Low contrast.



- High contrast image
- Histogram is (somewhat) uniformly distributed

In case of colored image, the distribution of the three color channels are calculated separately.

BVT can be better.

How can we Do better?

- how can contrast be further increased?

** Previously we increased contrast by
"Contrast Stretching".

** Contrast can be further increased
by "Histogram Equalization"

Histogram Equalization

Why Histogram Equalization?

→ To increase entropy

→ Entropy: $I = \sum_{r_k} -f(r_k) \log(f(r_k))$

- entropy দিবে একটি image এ
কত information আছে তা জানা

যায়।

- Histogram কতটা ছড়ান আছে।

- entropy যত বেশি info তত বেশি

- একটি image এর uncertainty
যত বেশি তার info তত বেশি

↳ entropy এই uncertainty
calculate করে।

↳ entropy বেশি হলে info বেশি

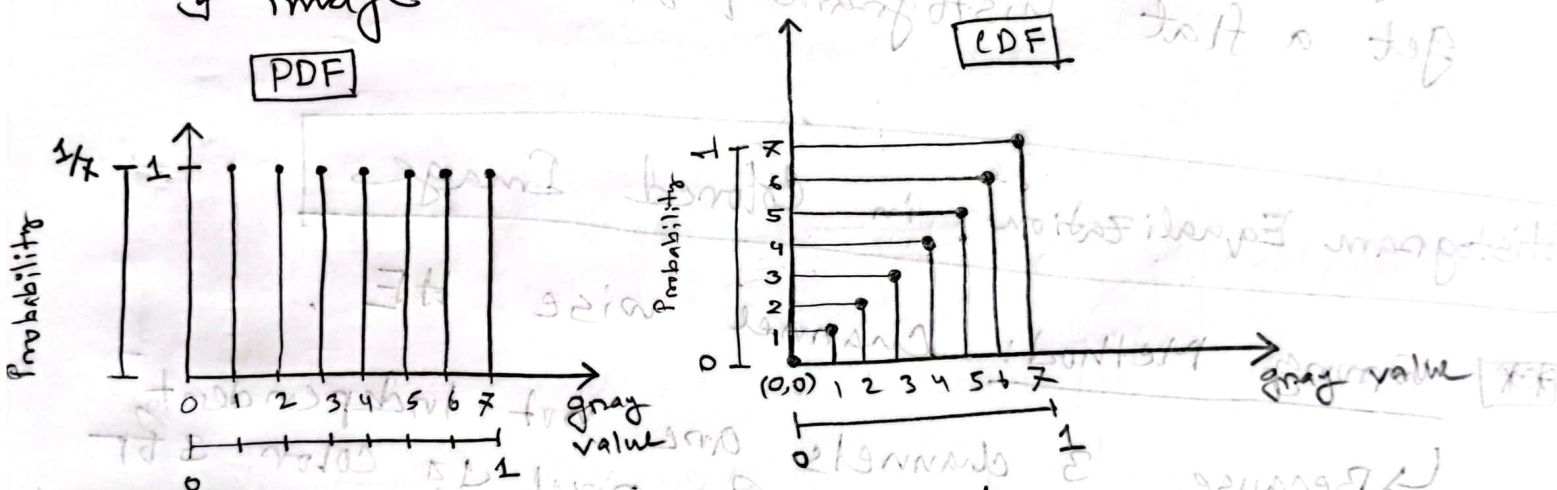
↳ একটি image এর uncertainty
info ত বেশি।

** যদি image এর Histogram uniformly distributed
যদি image এর uncertainty তত বেশি। কারণ
প্রত্যেকটি gray scale এর value এর probability
same থাকে। তাই এই image এর info অনেক
বেশি।

So to sum up,

Entropy মত বৈকি, Uncertainty তত বৈকি।
 Uncertainty মত বৈকি image এর into তত
 বৈকি & Histogram তত বৈকি Uniformly
 distributed.

Histogram মত বৈকি Uniformly distributed
 & image এর contrast তত বৈকি।



- PDF is flat, $f(r_k) = \frac{1}{L-1}$; $L = \text{Quantization level}$
- CDF is linear between $[\min, \max]$, slope = $\frac{1}{L-1}$

Observation

If we observe there should be $\gamma\%$ pixels with the gray scale value $\leq \gamma$.

Formula to Equalize Histogram

map x to its CDF value to get a flat histogram / linear CDF.

Histogram Equalization in Colored Image

Wrong Method: Channel wise HE.

↳ Because 3 channels are not independent to each others. One pixel has color 3 bits channel form has 24 bits. So 3 bits are independent equalize separately in 3 bits. $\leftarrow$
मरिउ contrast बरु. $\leftarrow$

Correct Method: Color space change कर. 1

$[0, 100]$ L
↓
black/white
Brightness into

$[-100, +100]$ A
↓
red/green
Color into

$[-100, +100]$ B
↓
blue/yellow

RGB to LAB

↓
only equalize 'L'

↓
The back to: LAB to RGB

Failure of Histogram Equalization

- ① Details in low or high intensity value is lost.
- ② Noise amplifies

AHE - Adaptive Histogram Equalization

✖✖ if image contains regions that are significantly lighter or darker, the contrast in those regions will not be sufficiently enhanced by Histogram Equalization

✖✖ The solution is to apply Histogram Equalization on a local neighbourhood

AHE Failure

** যেখানে intensity এর variation কম
তখন HE & AHE fail করে।

CLAHE - Contrast Limited Adaptive Histogram Equalization

** while applying AHE, particular
gray scale এর pixel count একটি
certain number এর বেশি হলে
একে বিচ্ছিন্ন করে distribute করে।